Statistical Mechanics: Exercises

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November 4, 2026

Scope of the exercises. The exercises are selected from three problem collections: A. A. Schekochihin and J. Devriendt, *Oxford Problem Sets 5–8: Statistical Mechanics* (2017), cited as [S]; D. Cohen, *Exercises in Statistical Mechanics* (Ben-Gurion University, 2019), cited as [C]; and D. A. R. Dalvit, J. Frastai and I. D. Lawrie, *Problems on Statistical Mechanics* (IOP, 1999), cited as [D]. Only problems that the note covers and that require more than a substitution into one formula are included. The problems are rewritten in the notation of the note, and some are merged or extended. Problems on quantum gases, kinetic theory, interacting lattice models and continuous transitions are left out, because the note does not cover them yet. The collections contain no problems on binary phase diagrams, so Section 7 of the note has no exercises here. Each exercise names the related sections of the note and the source, and ends with an answer for checking. Section 7 of this sheet adds guided exercises on point defects, built from the equations of a review article (see the start of that section).

Units and notation. The Boltzmann constant k_B is kept explicit, and $\beta = 1/(k_B T)$. Where a problem needs the classical ideal gas, the canonical partition function of N identical particles of mass m in the volume V is given as $Z_N = V^N/(N!\,\lambda_{\mathrm{th}}^{3N})$, with the thermal wavelength $\lambda_{\mathrm{th}} = h/\sqrt{2\pi m k_B T}$.

1 Entropy Maximization

Exercise 1: Loaded die. A die has the outcomes $i = 1, \ldots, 6$ with unknown probabilities p_i.

(a) With no information beyond normalization, find the MaxEnt distribution and the expected outcome $\langle i \rangle$.

(b) Many throws give the average $\bar{\imath} = 3.667$. Use MaxEnt with the constraint $\langle i \rangle = \bar{\imath}$ to show that $p_i = x^i / \sum_{j=1}^{6} x^j$, with $x = e^{-\lambda}$ for the multiplier λ, and that x solves

$$\sum_{i=1}^{6} (i - \bar{\imath})\, x^i = 0.$$

(c) Solve for x numerically, list the p_i, and compare the entropy with $\ln 6$.

Answer. (a) $p_i = 1/6$ and $\langle i \rangle = 3.5$. (c) $x = 1.059$ ($\lambda = -0.057$), and $p_1, \ldots, p_6 = 0.144$, 0.152, 0.161, 0.171, 0.181, 0.192. The entropy is 1.787, only 0.005 below $\ln 6 = 1.792$, because the constraint is weak.
Related: Sections 1.2 and 3 of the note. *Source:* [S] 5.3.

Exercise 2: Additivity forces the logarithm. Suppose that the entropy of an isolated system is some function $S = \phi(\Omega)$ of the number of accessible microstates, with ϕ increasing.

(a) Two independent systems have Ω_1 and Ω_2 microstates. Show that additivity of the entropy requires $\phi(\Omega_1 \Omega_2) = \phi(\Omega_1) + \phi(\Omega_2)$, and hence $\phi(1) = 0$ and $\phi(\Omega^n) = n\, \phi(\Omega)$ for every positive integer n.

(b) For integers $\Omega, M \geq 2$ and a large integer n, choose the integer m with $M^m \leq \Omega^n < M^{m+1}$. Use the monotonicity of ϕ to show that

$$\left| \frac{\phi(\Omega)}{\phi(M)} - \frac{\ln \Omega}{\ln M} \right| \leq \frac{1}{n},$$

and conclude that $\phi(\Omega) = k \ln \Omega$ with a constant $k > 0$.

(c) Show that the Gibbs–Shannon entropy of a uniform distribution over Ω microstates has this form, and explain why the note sets $k = k_B$.

Answer. (b) Both $\phi(\Omega)/\phi(M)$ and $\ln \Omega / \ln M$ lie in $[m/n, (m+1)/n)$. The ratio $k = \phi(M)/\ln M$ is therefore the same for every M. (c) $S = \ln \Omega$ for $p_i = 1/\Omega$, and k_B only fixes the units.
Related: Sections 1.1 and 2 of the note. *Source:* [C] 1043.

2 The Microcanonical Ensemble

Exercise 3: Two-level system and negative temperature. N distinguishable, non-interacting particles each have the energy $-\varepsilon$ or $+\varepsilon$. The system is isolated with the energy E, so that $n = (N + E/\varepsilon)/2$ particles occupy the upper level.

(a) Find $S(E)$ for $N, n \gg 1$, and sketch S against E.

(b) From $1/T = \partial S/\partial E$, find $T(E)$. Show that $T < 0$ for $E > 0$, and explain why such states are hotter than any state with $T > 0$.

(c) The system at $T < 0$ touches a heat bath at $T_0 > 0$. Use the entropy change $\delta S = k_B(\beta_1 - \beta_2)\,\delta U_1$ of Section 3.4 of the note to find the direction of the energy flow.

(d) Find the entropy change from $T = 0^+$ to $T \to \pm\infty$ and express it in bits.

(e) Explain why an ideal gas in a box cannot reach $T < 0$.

Answer. (a) $S = -Nk_B[x \ln x + (1-x)\ln(1-x)]$ with $x = n/N$, maximal at $E = 0$. (b) $\beta = (2\varepsilon)^{-1}\ln[(N-n)/n] = -(2\varepsilon)^{-1}\ln[(N\varepsilon+E)/(N\varepsilon-E)]$. As E increases, β decreases continuously from $+\infty$ through 0 to $-\infty$. (c) Energy flows from the system to the bath, because the bath has the larger β. (d) $\Delta S = Nk_B \ln 2$, that is, N bits. (e) The kinetic energy has no upper bound, so $S(E)$ increases for all E.
Related: Sections 2 and 3.4 of the note. *Source:* [C] 1047; [D] 3.6.

Exercise 4: Frenkel defects. A crystal has N atoms on N lattice sites and M interstitial sites, with M of order N. Moving an atom from a lattice site to an interstitial site costs the energy ω. Let $x = e^{-\beta\omega}$.

(a) Show that the canonical partition function is

$$Z(x) = \sum_n \binom{N}{n}\binom{M}{n} x^n.$$

(b) Treat n as continuous, with $d\ln n!/dn \approx \ln n$, and find the largest term. Show that $n^2/[(N-n)(M-n)] = x$, and hence $\bar{n} \approx \sqrt{NM}\, e^{-\beta\omega/2}$ for $x \ll 1$.

(c) Interpret the condition of part (b) as a law of mass action for the reaction "lattice atom $\rightleftharpoons$ vacancy + interstitial".

(d) Keep only the largest term in $\ln Z$ and find F, U, S and C_V for $x \ll 1$.

(e) Explain why a Gaussian sum around the largest term changes $\ln Z$ only by $O(\ln \bar{n})$.

Answer. (b) For $N = M = 4000$ and $x = e^{-12}$, the exact mean is 9.64 and the approximation gives 9.92. (d) $\ln Z \approx 2\bar{n}$, so $F = -2k_B T\bar{n}$, $U = \bar{n}\omega$, $S = k_B\bar{n}(2 + \beta\omega)$ and $C_V = 2k_B\bar{n}(\beta\omega/2)^2$. The factor $1/2$ in the exponent of $\bar{n}$ reflects that each defect creates two independent objects, a vacancy and an interstitial.
Related: Sections 2, 3 and 3.3 of the note. *Source:* [C] 2311.

3 The Canonical Ensemble

Exercise 5: Elastic chain and superfluous constraints. Here, τ denotes a tension and Λ the length of a chain. A one-dimensional chain has N segments. Each segment is horizontal, with the length a and the energy 0, or vertical, with the length 0 and the energy τa, where τ is a fixed tension.

(a) Find the canonical partition function and the mean energy $U(T)$.

(b) Show that the mean length is $\langle\Lambda\rangle = Na - U/\tau$. For $\beta\tau a \ll 1$, show that $\tau = Ak_B T(\langle\Lambda\rangle - \Lambda_0)$, a Hooke law, and find A and Λ_0. Explain why a hotter chain needs a larger tension for the same length.

(c) Find the heat capacity at fixed τ, and explain why it vanishes at low and at high temperature.

(d) Find the condition on U for $T < 0$.

(e) Maximize the entropy with two constraints, $\langle E\rangle = U$ and $\langle\Lambda\rangle = \Lambda$, with the multipliers β and σ. Show that a solution exists only if $\Lambda = Na - U/\tau$, and that only the combination $\beta - \sigma/\tau$ is determined. Relate this result to the uniqueness condition for the multipliers in Section 5 of the note.

Answer. (a) $Z = (1 + e^{-\beta\tau a})^N$ and $U = N\tau a/(e^{\beta\tau a} + 1)$. (b) $\Lambda_0 = Na/2$ and $A = 4/(Na^2)$. The tension opposes the entropy, which favors the disordered half-and-half state, and the entropic restoring force grows with T. (c) $C = Nk_B(\beta\tau a)^2 e^{\beta\tau a}/(e^{\beta\tau a} + 1)^2$, the Schottky form, with a maximum near $k_B T \approx 0.42\,\tau a$. (d) $U > N\tau a/2$. (e) Every microstate satisfies $E + \tau\Lambda = N\tau a$, so the covariance matrix of (E, Λ) is singular.
Related: Sections 1.2, 3 and 5 of the note. *Source:* [S] 5.5.

Exercise 6: Zipper model and the thermodynamic limit. A double-stranded molecule has N links that open only in sequence from one end, and the last link never opens. An open link costs the energy ε and has g orientations. A state with p open links has the energy $p\varepsilon$ and the degeneracy g^p.

(a) With $x = g\,e^{-\beta\varepsilon}$, show that $Z = (1 - x^N)/(1 - x)$.

(b) Find $\langle p\rangle$, and show that $\langle p\rangle/N \to 0$ for $x < 1$ and $\langle p\rangle/N \to 1$ for $x > 1$ as $N \to \infty$.

(c) Show that $\langle p\rangle \approx (N-1)/2 + (x-1)(N^2-1)/12$ near $x = 1$, and find the transition temperature T_c.

(d) Find the heat capacity at $x = 1$ and show that it grows as N^2.

(e) Explain why the free energy is analytic in T for every finite N, and why a transition nevertheless appears in the limit. Which order is the transition?

Answer. (b) $\langle p\rangle = x/(1-x) - Nx^N/(1-x^N)$. For $N = 500$, $\langle p\rangle/N = 0.04$, 0.50 and 0.96 at $x = 0.95$, 1 and 1.05. (c) $T_c = \varepsilon/(k_B \ln g)$. The slope is the variance $(N^2-1)/12$ of a uniform distribution. (d) $C = k_B(\ln g)^2(N^2-1)/12$. (e) Z is a polynomial in x, so $\ln Z$ is analytic for finite N. In the limit, the fraction of open links jumps from 0 to 1 with the latent heat ε per link, a first-order transition.
Related: Sections 3 and 6 of the note. *Source:* [C] 2360.

Exercise 7: Free energy of a perturbed Hamiltonian. A classical system has the Hamiltonian $\mathcal{H} = \mathcal{H}_0 + \lambda\mathcal{H}_1$. Let F_0 and $\langle\cdot\rangle_0$ denote the free energy and the averages at $\lambda = 0$.

(a) Show that $e^{-\beta(F-F_0)} = \langle e^{-\beta\lambda\mathcal{H}_1}\rangle_0$.

(b) Expand in λ and show that

$$F = F_0 + \lambda\langle\mathcal{H}_1\rangle_0 - \frac{\beta\lambda^2}{2}\,\mathrm{Var}_0(\mathcal{H}_1) + \frac{\beta^2\lambda^3}{6}\,\kappa_3 + O(\lambda^4),$$

where $\kappa_3 = \langle(\mathcal{H}_1 - \langle\mathcal{H}_1\rangle_0)^3\rangle_0$.

(c) Show that $\partial^2 F/\partial\lambda^2 = -\beta\,\mathrm{Var}_\lambda(\mathcal{H}_1) \le 0$ for every λ, and deduce the Gibbs–Bogoliubov bound $F \le F_0 + \lambda\langle\mathcal{H}_1\rangle_0$.

(d) Find $U = \langle\mathcal{H}\rangle$ to first order in λ.

Answer. (b) The expansion is the cumulant expansion of $\ln\langle e^{-\beta\lambda\mathcal{H}_1}\rangle_0$. (c) $F(\lambda)$ is concave, so the curve lies below the tangent at $\lambda = 0$. (d) $U = U_0 + \lambda\,[\langle\mathcal{H}_1\rangle_0 - \beta\,\mathrm{Cov}_0(\mathcal{H}_0, \mathcal{H}_1)] + O(\lambda^2)$.
Related: Sections 3 and 3.3 of the note. *Source:* [D] 3.13.

4 Thermodynamic Potentials and Legendre Transformations

Exercise 8: Heat capacities and compressibilities. Here, $\alpha_P = V^{-1}(\partial V/\partial T)_P$ is the thermal expansivity, $\kappa_T = -V^{-1}(\partial V/\partial P)_T$ the isothermal compressibility and $\kappa_S = -V^{-1}(\partial V/\partial P)_S$ the adiabatic compressibility, all at fixed N.

(a) From the equality of the mixed second derivatives of $F(T, V, N)$ and of $G(T, P, N)$, derive the Maxwell relations $(\partial S/\partial V)_T = (\partial P/\partial T)_V$ and $(\partial S/\partial P)_T = -(\partial V/\partial T)_P$.

(b) Show that $T\,dS = C_V\,dT + T(\alpha_P/\kappa_T)\,dV$ and $T\,dS = C_P\,dT - TV\alpha_P\,dP$.

(c) Prove that

$$C_P - C_V = \frac{TV\alpha_P^2}{\kappa_T}, \qquad \frac{C_P}{C_V} = \frac{\kappa_T}{\kappa_S}.$$

(d) Express C_P and C_V in terms of T, V, α_P, κ_T and κ_S.

(e) Use the stability conditions of the note to show that $C_P \geq C_V > 0$ and $\kappa_T \geq \kappa_S > 0$.

Answer. (b) $(\partial P/\partial T)_V = \alpha_P/\kappa_T$ by the cyclic rule. (d) $C_P = TV\alpha_P^2/(\kappa_T - \kappa_S)$ and $C_V = TV\alpha_P^2\kappa_S/[\kappa_T(\kappa_T - \kappa_S)]$. (e) $C_V > 0$ and $\kappa_T > 0$ follow from the concavity of S and the convexity of F in V. Then $C_P - C_V \geq 0$, and $\kappa_S = \kappa_T C_V/C_P$.
Related: Sections 3.3, 4.4 and 5.3 of the note. *Source:* [D] 2.4.

Exercise 9: Rubber band in the constant-tension ensemble. A polymer has N monomers on a line. Each monomer has the length a and the energy E_a, or the length b and the energy E_b, with $a > b$. The polymer is held at the temperature T under the tension τ.

(a) Maximize the entropy with the constraints $\langle E\rangle$ and $\langle\Lambda\rangle$, identify the multiplier of the length as $-\beta\tau$, and show that

$$\Gamma(T, \tau) = \left[e^{-\beta(E_a - \tau a)} + e^{-\beta(E_b - \tau b)}\right]^N.$$

(b) Show that $G(N, \tau, T) = -k_B T \ln\Gamma = F - \tau\Lambda$ is the Legendre transform of $F(N, \Lambda, T)$ in Λ. Name the ensemble by the natural variables of G, and find $\Lambda = -\partial G/\partial\tau$.

(c) Find $\Lambda(T \to \infty, \tau)$. For $E_a = E_b$ and weak tension, show that $\tau = 4k_BT(\Lambda - \Lambda_\infty)/[N(a-b)^2]$.

(d) At $\tau = 0$ and $E_a = E_b$, find the distribution $P(\Lambda)$ of the length for large N. Show that the entropic force $\tau = -k_BT\,\partial \ln P/\partial\Lambda$ reproduces part (c).

(e) Let each monomer take any length in $[b, a]$ with equal energy. Find Γ and $\Lambda(T, \tau)$.

Answer. (b) The $N\tau T$ ensemble, with $\Lambda = N\,[b + (a-b)\,p_a]$ and $p_a = \{1 + e^{\beta[(E_a - \tau a) - (E_b - \tau b)]}\}^{-1}$. (c) $\Lambda_\infty = N(a+b)/2$. (d) $P(\Lambda)$ is Gaussian with the mean Λ_∞ and the variance $N(a-b)^2/4$. (e) $\Gamma = \left[(e^{\beta\tau a} - e^{\beta\tau b})/(\beta\tau(a-b))\right]^N$ and $\Lambda = N\left[(ae^{\beta\tau a} - be^{\beta\tau b})/(e^{\beta\tau a} - e^{\beta\tau b}) - k_BT/\tau\right]$.
Related: Sections 1.2, 4.4 and 4.5 of the note. *Source:* [C] 2351.

5 The Grand Canonical Ensemble

Exercise 10: The grand canonical distribution from a reservoir. A small system exchanges energy and particles with a large reservoir, and the two together are isolated, with the total energy E_{tot} and the total particle number N_{tot}.

(a) Assume equal probabilities for the microstates of the isolated whole. Show that the probability of a system microstate i is proportional to $\Omega_R(E_{\text{tot}} - E_i, N_{\text{tot}} - N_i)$.

(b) Expand $S_R = k_B \ln \Omega_R$ to first order and recover $p_i \propto e^{-\beta(E_i - \mu N_i)}$. Identify T and μ with derivatives of S_R.

(c) Estimate the second-order terms and show that they vanish as the reservoir becomes infinitely large.

(d) Compare this derivation with the MaxEnt derivation in Section 5 of the note. Which assumption replaces the constraints on $\langle E \rangle$ and $\langle N \rangle$?

Answer. (b) $1/T = \partial S_R/\partial E$ and $-\mu/T = \partial S_R/\partial N$. (c) The second derivatives scale as $1/N_R$, so their terms are of order E_i^2/N_R. (d) The reservoir fixes T and μ through the reservoir entropy, whereas MaxEnt fixes the multipliers through the measured averages. Both give the same distribution.
Related: Sections 3.4 and 5 of the note. *Source:* [S] 6.11.

Exercise 11: Energy fluctuations with particle exchange. Here, $z = e^{\beta\mu}$ denotes the fugacity.

(a) From $\ln \Xi(\beta, z)$ at fixed V, show that $\mathrm{Var}_G(E) = -(\partial U/\partial\beta)_z$, $\mathrm{Var}(N) = z(\partial N/\partial z)_\beta$, and $\mathrm{Cov}(E, N) = z(\partial U/\partial z)_\beta = -(\partial N/\partial\beta)_z$.

(b) In the thermodynamic limit, U in the grand canonical ensemble equals $U(\beta, V, N)$ in the canonical ensemble with $N = N(\beta, z)$. Use the chain rule to prove

$$\mathrm{Var}_G(E) = \mathrm{Var}_C(E) + \left(\frac{\partial U}{\partial N}\right)^2_{T,V} \mathrm{Var}(N).$$

(c) Explain why the grand canonical energy fluctuation exceeds the canonical one.

(d) For the monatomic ideal gas, with $U = \frac{3}{2}Nk_BT$ and $\mathrm{Var}(N) = N$, find the ratio $\mathrm{Var}_G(E)/\mathrm{Var}_C(E)$.

Answer. (b) $(\partial U/\partial\beta)_z = (\partial U/\partial\beta)_N + (\partial U/\partial N)_\beta(\partial N/\partial\beta)_z$, and $-(\partial N/\partial\beta)_z = (\partial U/\partial N)_\beta \,\mathrm{Var}(N)$. (c) The exchanged particles carry energy. (d) $\mathrm{Var}_G(E) = \frac{15}{4}N(k_BT)^2$ and $\mathrm{Var}_C(E) = \frac{3}{2}N(k_BT)^2$, so the ratio is $5/2$.
Related: Sections 3.3, 5 and 5.1 of the note. *Source:* [D] 3.28.

Exercise 12: Particle-number fluctuations and compressibility.

(a) Use the Gibbs–Duhem relation at fixed T, $dP = (N/V)\,d\mu$, to show that $(\partial N/\partial\mu)_{T,V} = N^2\kappa_T/V$. Hence express $\mathrm{Var}(N)/N$ through κ_T.

(b) Show that the third cumulant is $\langle (N - \langle N\rangle)^3\rangle = (k_BT)^2(\partial^2 N/\partial\mu^2)_{T,V}$.

(c) For the classical ideal gas, show that $\Xi = \exp(zV/\lambda_{\mathrm{th}}^3)$, that all cumulants of N equal $\langle N\rangle$, and that N has the Poisson distribution $P(N) = \langle N\rangle^N e^{-\langle N\rangle}/N!$.

(d) Explain why $\mathrm{Var}(N)/N$ grows without bound as a fluid approaches the critical point.

Answer. (a) $\mathrm{Var}(N)/N = k_BT\,(N/V)\,\kappa_T$. (b) The nth cumulant is $(k_BT)^{n-1}\partial^{n-1}N/\partial\mu^{n-1}$. (c) $N = zV/\lambda_{\mathrm{th}}^3 \propto e^{\beta\mu}$, so every derivative $k_BT\,\partial/\partial\mu$ returns N. (d) $\kappa_T \to \infty$ at the critical point, where $\partial P/\partial v = 0$.
Related: Sections 5.1 and 5.7 of the note. *Source:* [D] 2.5; [C] 1027; [S] 6.9.

Exercise 13: Langmuir adsorption. A classical ideal gas at the pressure P and the temperature T is in contact with a wall that has N_0 independent sites. Each site is empty, with the energy 0, or holds one molecule, with the energy $-\varepsilon$.

(a) Use the result of Exercise 12 to show that the fugacity of the gas is $z = P\lambda_{\text{th}}^3/(k_BT)$.

(b) Treat the sites as a grand canonical system with the same T and μ as the gas. Find the grand partition function of one site and the mean number of adsorbed molecules $\langle N_s \rangle$.

(c) Write the coverage $\theta = \langle N_s \rangle / N_0$ as $P/(P + P_0(T))$ and find $P_0(T)$. Discuss the limits of low and high pressure.

(d) Find $\text{Var}(N_s)$, and explain why it vanishes at full coverage.

Answer. (b) $\xi = 1 + ze^{\beta\varepsilon}$ and $\langle N_s \rangle = N_0 ze^{\beta\varepsilon}/(1 + ze^{\beta\varepsilon})$. (c) $P_0 = (k_BT/\lambda_{\text{th}}^3)\, e^{-\beta\varepsilon}$. At low pressure $\theta \approx P/P_0$ (Henry's law), and at high pressure $\theta \to 1$. (d) $\text{Var}(N_s) = N_0\theta(1-\theta)$.
Related: Sections 5 and 5.5 of the note. *Source:* [D] 3.25.

Exercise 14: The NPT ensemble of an ideal gas. The partition function of the NPT ensemble in the note is $\Gamma = \sum_i e^{-\beta(E_i + PV_i)}$. For a gas in a container of variable volume, replace the sum over volumes by $\int_0^\infty dV/\Delta V$, where ΔV is a "quantum of volume".

(a) With $Z_N(V)$ of the classical ideal gas, show that

$$\Gamma = \frac{1}{\Delta V\, \lambda_{\text{th}}^{3N}} \left(\frac{k_BT}{P} \right)^{N+1}.$$

(b) Find $G = -k_BT \ln \Gamma$. Show that $G = N\mu$ with $\mu = k_BT \ln(P\lambda_{\text{th}}^3/k_BT)$, up to a term that does not grow with N, and find the condition on ΔV for that term to be negligible.

(c) Find $\langle V \rangle = (\partial G/\partial P)_{T,N}$ and the equation of state.

(d) Find $\text{Var}(V)/\langle V \rangle^2$, and relate it to κ_T.

Answer. (b) $G = Nk_BT \ln(P\lambda_{\text{th}}^3/k_BT) - k_BT \ln(k_BT/P\Delta V)$. The second term is negligible if $|\ln(\langle V \rangle / N\Delta V)| \ll N$, which holds for any physically reasonable ΔV. (c) $\langle V \rangle = (N+1)k_BT/P \approx Nk_BT/P$. (d) $\text{Var}(V) = k_BT\langle V \rangle \kappa_T$, so $\text{Var}(V)/\langle V \rangle^2 = 1/(N+1)$.
Related: Sections 5.3 and 5.4 of the note. *Source:* [S] 6.12.

Exercise 15: Mixing gases at different temperatures. The monatomic classical ideal gas has the entropy

$$S(U, V, N) = Nk_B \left[\ln\left(\frac{V}{N} \left(\frac{4\pi m U}{3Nh^2} \right)^{3/2} \right) + \frac{5}{2} \right].$$

Two such gases, with equal particle masses, occupy the volumes V_1 and V_2, with the particle numbers N_1 and N_2 and the temperatures T_1 and T_2. The wall between them is removed, and the combined system comes to equilibrium in $V = V_1 + V_2$.

(a) Find the final temperature.

(b) Find ΔS when the two gases are identical and when they are different species.

(c) Show that the difference between the two results is

$$\Delta S_{\text{diff}} - \Delta S_{\text{ident}} = -Nk_B \left[x_1 \ln x_1 + x_2 \ln x_2\right], \qquad x_i = N_i/N,$$

independent of T_1, T_2, V_1 and V_2.

(d) Relate the result of part (c) to $\Delta S_{\text{mix}} \geq 0$ in Section 5.6 of the note and to the ideal-solution model of Section 7.7.

Answer. (a) $T_f = (N_1T_1 + N_2T_2)/N$. (b)

$$\Delta S_{\text{ident}} = \tfrac{3}{2}k_B \left[N \ln T_f - N_1 \ln T_1 - N_2 \ln T_2\right] + k_B \left[N \ln \frac{V}{N} - N_1 \ln \frac{V_1}{N_1} - N_2 \ln \frac{V_2}{N_2}\right],$$

and ΔS_{diff} has $k_B[N_1 \ln(V/V_1) + N_2 \ln(V/V_2)]$ as the second bracket. (d) The difference is the ideal mixing entropy, which the ideal-solution model assumes for the composition dependence of g. *Related:* Sections 5.6 and 7.7 of the note. *Source:* [C] 1045; [S] 5.6.

6 First-Order Phase Transitions and the Unary System

Exercise 16: Critical point of a model fluid.

(a) Below T_c, a model isotherm $P(v)$ has a region with $\partial P/\partial v > 0$. Use the convexity of $f(v)$ in Section 6.3 of the note to explain why this region is unstable. Show that the region shrinks to one point at T_c, so that the critical point satisfies

$$\left(\frac{\partial P}{\partial v}\right)_T = 0, \qquad \left(\frac{\partial^2 P}{\partial v^2}\right)_T = 0.$$

(b) The Dieterici equation of state per particle is $P(v-b) = k_BT\, e^{-a/(k_BTv)}$. Find T_c, v_c and P_c.

(c) In the reduced variables $\tilde{P} = P/P_c$, $\tilde{T} = T/T_c$ and $\tilde{v} = v/v_c$, show that $\tilde{P}(2\tilde{v} - 1) = \tilde{T} \exp[2(1 - 1/\tilde{T}\tilde{v})]$, a law of corresponding states.

(d) Compare the critical ratio $P_c v_c / k_B T_c$ with the van der Waals value $3/8$ and with the measured values of about 0.29 for simple gases.

Answer. (a) $\partial P/\partial v = -\partial^2 f/\partial v^2$, which must be ≤ 0. At T_c, the minimum and the maximum of $P(v)$ merge into an inflection point. (b) $T_c = a/(4k_B b)$, $v_c = 2b$ and $P_c = a/(4e^2 b^2)$. (d) $P_c v_c / k_B T_c = 2/e^2 = 0.271$.
Related: Section 6.3 of the note. *Source:* [S] 8.2.

Exercise 17: Van der Waals coexistence near the critical point. The reduced van der Waals equation of state is $\tilde{P} = 8\tilde{T}/(3\tilde{v} - 1) - 3/\tilde{v}^2$.

(a) Show that the critical isotherm satisfies $\tilde{P} - 1 \approx -\frac{3}{2}(\tilde{v} - 1)^3$.

(b) On the critical isochore $\tilde{v} = 1$ above T_c, show that $P_c \kappa_T = [6(\tilde{T} - 1)]^{-1}$.

(c) Below T_c, apply Maxwell's equal-area rule from Section 6.3 of the note to show that $\tilde{v}_g - \tilde{v}_l \approx 4(1 - \tilde{T})^{1/2}$.

(d) Use the Clapeyron equation to show that the latent heat per particle vanishes as $\ell \approx 6 k_B T_c (1 - \tilde{T})^{1/2}$.

Answer. (c) A numerical Maxwell construction gives $(\tilde{v}_g - \tilde{v}_l)/(1 - \tilde{T})^{1/2} = 4.12$, 4.01 and 4.001 for $1 - \tilde{T} = 10^{-2}$, 10^{-3} and 10^{-4}. (d) $d\tilde{P}/d\tilde{T} \to 4$ on the coexistence curve at T_c, and $P_c v_c = \frac{3}{8} k_B T_c$. The latent heat and Δv vanish together at T_c, as Section 6.3 of the note states.
Related: Sections 6.2 and 6.3 of the note. *Source:* [D] 5.8.

Exercise 18: Latent heat along the vaporization curve. A liquid is incompressible and does not expand with temperature. The vapor volume per particle v_g is much larger than v_l.

(a) Differentiate $\ell = T(s_g - s_l)$ along the coexistence curve. Use a Maxwell relation and the Clapeyron equation to show that

$$\frac{d\ell}{dT} = \frac{\ell}{T} + \Delta c_P - \frac{\ell}{v_g}\left(\frac{\partial v_g}{\partial T}\right)_P, \qquad \Delta c_P = c_{P,g} - c_{P,l}.$$

(b) For an ideal-gas vapor and a constant Δc_P, show that $\ell = \ell_0 + \Delta c_P \, T$.

(c) A saturated vapor expands adiabatically. Show that some liquid condenses if

$$c_{P,l} + T \, \frac{d}{dT}\left(\frac{\ell}{T}\right) < 0.$$

(d) Check the condition for water at 100 °C, with $c_{P,l} = 75.3$, $c_{P,g} = 33.6$ J mol^{-1} K^{-1} and $\ell = 40.7$ kJ mol^{-1}.

Answer. (a) $(\partial s/\partial P)_T = -(\partial v/\partial T)_P$ and $dP/dT = \ell/(Tv_g)$. (c) The entropy of the saturated vapor along the curve is $s_l + \ell/T$, with $ds_l/dT = c_{P,l}/T$. If this entropy decreases with T, an adiabat that starts on the curve enters the two-phase region on cooling. (d) The left side equals -75 J mol^{-1} K^{-1}, so expanding water vapor condenses, as in cloud formation.
Related: Sections 4.5, 6.2 and 6.3 of the note. *Source:* [S] 8.8.

Exercise 19: Boiling point and altitude. An isothermal atmosphere at $T_A = 300.15$ K consists of an ideal gas with the molar mass 30 g mol^{-1}, and the pressure at sea level is P_0. A liquid with the molar latent heat 1000 cal mol^{-1} boils at 105 °C at sea level and at 95 °C on the top of a mountain.

(a) Use the Boltzmann distribution of a molecule in the gravitational potential mgh to find $P_A(h)$.

(b) Integrate the Clausius–Clapeyron equation to find the vapor pressure of the liquid $P_{\text{vap}}(T)$, with $P_{\text{vap}}(378.15\text{ K}) = P_0$.

(c) Find the height of the mountain.

Answer. (a) $P_A = P_0 \exp(-Mgh/RT_A)$. (b) $P_{\text{vap}} = P_0 \exp[-(\ell_m/R)(1/T - 1/378.15\text{ K})]$, with the molar latent heat ℓ_m. (c) $h = (\ell_m T_A/Mg)(1/368.15 - 1/378.15)$ K^{-1} $= 307$ m.
Related: Sections 3 and 6.3 of the note. *Source:* [C] 5955.

Exercise 20: Cooling by solidification in helium-3. Below about 1 K, the entropy per particle of liquid ^{3}He is $s_l = (\pi^2/2)k_B T/T_F$ with $T_F \approx 5$ K. In the solid, the nuclear spins 1/2 are disordered and independent, so that $s_s = k_B \ln 2$. The solid is denser, and $\Delta v = v_l - v_s > 0$ is independent of T.

(a) Find the temperature T^* at which the two phases have equal entropy.

(b) Use the Clapeyron equation to show that the melting curve is

$$P(T) = P^* + \frac{\pi^2 k_B}{4T_F\,\Delta v}\,(T - T^*)^2,$$

with a minimum at T^*. Explain why the liquid is the low-entropy phase below T^*.

(c) State the order of the transition and find the jump in the heat capacity per particle.

(d) The liquid at $T_0 < T^*$ is compressed adiabatically. Assume that T_F does not depend on the pressure. Show that T stays at T_0 until the melting curve is reached.

(e) Further compression converts liquid to solid at fixed total entropy. Show that the system moves down the melting curve to lower T, with the solid fraction $n_s(T) = (T_0 - T)/(T^* - T)$.

Answer. (a) $T^* = (2\ln 2/\pi^2)\,T_F = 0.70$ K for $T_F = 5$ K (measured: 0.32 K). (b) $s_l - s_s = (\pi^2 k_B/2T_F)(T - T^*)$. (c) First order, with $\Delta c = (\pi^2/2)k_B T/T_F$ and $c_s = 0$. (e) The solid has the larger entropy below T^*, so freezing at fixed S absorbs heat from the liquid. This effect is Pomeranchuk cooling.
Related: Sections 6.2 and 6.3 of the note. *Source:* [C] 4552.

7 Point Defects in Crystals

Purpose and style of this section. Exercises 21–33 turn the equations of C. Freysoldt *et al.*, "First-principles calculations for point defects in solids", Rev. Mod. Phys. **86**, 253 (2014), cited as [F], into guided derivations. Each part is a question that leads to the next step, and the exercise ends at a numbered equation of [F]. Most questions use only Sections 1–7 of the note. Exercises 25, 26 and 29 introduce the Fermi–Dirac and Bose–Einstein distributions, and Exercise 32 introduces diffusion. Both topics lie beyond the note, so these exercises build the needed results from the canonical and grand canonical ensembles.

Notation for point defects. In this section, X^q denotes a defect X in the charge state q, E^f and G^f its formation energy and formation Gibbs energy, and $c = n/N$ its concentration per lattice site. The internal degeneracy of a defect is g_X, written g in [F], so that g keeps its meaning of the Gibbs free energy per particle. The Fermi level E_F is the electron chemical potential, measured from the valence-band maximum (VBM). The energy $e^2/(4\pi\epsilon_0) = 14.40$ eV Å is used for Coulomb estimates.

Exercise 21: The formation energy as a grand potential. A supercell of a crystal contains one defect X^q. The crystal exchanges atoms of species i with reservoirs at the chemical potentials μ_i and exchanges electrons with a reservoir at the chemical potential E_F.

(a) Which variables do the reservoirs fix, and which extensive quantities of the crystal therefore

fluctuate? Which Legendre transform of U in Section 5 of the note has these fixed variables as natural variables?

(b) To create the defect, n_i atoms of species i enter the supercell ($n_i < 0$ if they leave). By how much does the free energy of the reservoirs change? Why is the right energy per atom μ_i and not the energy of an isolated atom?

(c) A charge state $q > 0$ means that q electrons have left the crystal for the electron reservoir. What energy does the reservoir gain? Combine parts (b) and (c) at $T = 0$ to obtain Eq. (1) of [F] without $E_{\rm corr}$:

$$E^f[X^q] = E_{\rm tot}[X^q] - E_{\rm tot}[\text{bulk}] - \sum_i n_i\mu_i + qE_F.$$

(d) Why is E^f a straight line in E_F, and what sets its slope? For a defect with several charge states, which charge state does the crystal choose at a given E_F? Compare this rule with the lower envelope $\mu = \min_\alpha \mu_\alpha$ of Section 6.3 of the note.

(e) Two defects A and B can bind into a complex AB. Show that the binding energy $E_b = E^f[A] + E^f[B] - E^f[AB]$, Eq. (2) of [F], does not depend on any μ_i. Why can a positive E_b still give few complexes in equilibrium? (Hint: compare the number of ways to place one complex with the number of ways to place two separate defects.)

Answer. (a) T, P, every μ_i and E_F are fixed, and the atom numbers and the electron number fluctuate. The potential is Φ generalized to several species and to electrons. (b) The reservoirs lose $\sum_i n_i\mu_i$, the free energy per atom of the reservoir phase. (c) The reservoir gains qE_F. (d) The slope is q, and the stable charge state has the lowest E^f, as in Fig. 1 of [F]. (e) Every n_i of AB equals the sum of those of A and B. Two separate defects have about N^2 placements, but a complex has only about zN, where z is the number of neighbor sites.
Related: Sections 4.2, 5.2, 5.5 and 6.3 of the note. *Source:* [F], Secs. I.B.1–2, Eqs. (1)–(2).

Exercise 22: Charge-state transition levels and the U parameter.

(a) At which Fermi level do the charge states q_1 and q_2 have equal formation energy? Derive Eq. (3) of [F]:

$$\varepsilon(q_1/q_2) = \frac{E^f(X^{q_1}; E_F = 0) - E^f(X^{q_2}; E_F = 0)}{q_2 - q_1}.$$

(b) For three charge states $+1$, 0 and -1, write $\varepsilon(+/0)$ and $\varepsilon(0/-)$ in terms of $E^f_q \equiv E^f(X^q; E_F = 0)$. Show that $U = E^f_{+1} + E^f_{-1} - 2E^f_0$, Eq. (43) of [F], equals $\varepsilon(0/-) - \varepsilon(+/0)$.

(c) If $U < 0$, is the neutral state ever the lowest line? At which Fermi level does the defect then switch directly from $+1$ to -1?

(d) In Section 6.3 of the note, a phase whose curve never touches the lower envelope is metastable everywhere. What plays the role of the temperature, and what the role of the phases, in the defect diagram?

(e) Why must $E^f(X^q)$ be computed with the atomic structure relaxed separately in each charge state for $\varepsilon(q_1/q_2)$ to be a thermodynamic level? (Exercise 31 contrasts this with optical levels.)

Answer. (a) Set $E^f(X^{q_1}) + q_1 E_F = E^f(X^{q_2}) + q_2 E_F$. (b) $\varepsilon(+/0) = E^f_0 - E^f_{+1}$ and $\varepsilon(0/-) = E^f_{-1} - E^f_0$. (c) No. The switch occurs at $\varepsilon(+/-) = [\varepsilon(+/0) + \varepsilon(0/-)]/2$. (d) E_F plays the role of T, and the charge states play the role of the phases. A transition level is a "melting point" of the charge state. (e) A thermodynamic level compares two equilibrium states, each at the minimum of its own energy surface.
Related: Sections 6 and 6.3 of the note. *Source:* [F], Secs. I.B.3 and II.D, Eqs. (3), (43).

Exercise 23: Configurational entropy and the equilibrium concentration. A crystal has N sites, and n identical defects occupy them. Each defect has g_X internal orientations.

(a) Why does the number of arrangements equal $W = (g_X N)!/[(g_X N - n)!\, n!]$? Which factorial removes the distinction between identical defects, and which approximation gives $W \approx (g_X N)^n/n!$ for $n \ll N$?

(b) Apply Stirling's formula to $S^{\mathrm{conf}} = k_B \ln W$ and show that, per site, $S^{\mathrm{conf}}(c) = k_B[c - c\ln c + c\ln g_X]$, Eq. (7) of [F].

(c) Compare this entropy with the ideal mixing entropy $-k_B[c\ln c + (1-c)\ln(1-c)]$ of Section 7.7 of the note. Which term does the dilute limit drop?

(d) The Gibbs energy per site is $G(c) = G[\mathrm{bulk}] + cG^f - TS^{\mathrm{conf}}(c)$, Eq. (39) of [F]. What does $\partial G/\partial c = 0$ give? Derive $c^{\mathrm{eq}} = g_X \exp(-G^f/k_BT)$, Eq. (41).

(e) Show that G at equilibrium is $G[\mathrm{bulk}] - k_BTc^{\mathrm{eq}}$, Eq. (40). Why does the crystal always gain free energy by forming some defects, however large G^f is? Look at the slope of S^{conf} at $c \to 0$.

(f) Define the defect chemical potential $\mu_X = \partial G/\partial c$. Why is $\mu_X = 0$ at equilibrium? Which conservation law that holds for atoms fails for vacancies, and which other "particles" of Exercise 26 share this property?

Answer. (b) $\ln W \approx n\ln(g_X N) - n\ln n + n$. (c) The expansion $-(1-c)\ln(1-c) \approx c$ gives the term c, and only terms of order c^2 are dropped. (d) $G^f + k_B T \ln(c/g_X) = 0$. (e) $dS^{\text{conf}}/dc = -k_B \ln(c/g_X) \to +\infty$ as $c \to 0$. (f) Vacancies form by moving atoms to the surface, so their number is not conserved. Phonons, with $\mu = 0$, behave in the same way.
Related: Sections 2, 4.5, 5.6 and 7.7 of the note. *Source:* [F], Secs. II.A.1 and II.B.3, Eqs. (4)–(8), (39)–(41).

Exercise 24: Arrhenius plots and their extrapolation. Table I of [F] lists, for the vacancy in Al, $H^f = 0.76$ eV and $S^f = 2.4\,k_B$ from experiment at the melting temperature $T^m = 933$ K, and DFT values at 0 K and at T^m.

(a) If H^f and S^f do not depend on T, what does a plot of $\ln c^{\text{eq}}$ against $1/T$ look like? Which feature of the plot gives H^f, and which gives S^f? This is Eq. (42) of [F].

(b) Evaluate $c^{\text{eq}}(T^m)$ with $g_X = 1$. By what factor would c^{eq} be smaller if the vibrational formation entropy were zero?

(c) Use $\partial^2 G^f/\partial T^2 = -C_P^f/T$, with the formation heat capacity C_P^f, as in Section 4 of the note. If $C_P^f > 0$, is $G^f(T)$ concave or convex?

(d) An experiment measures G^f only near T^m and extrapolates the tangent line to $T = 0$. Show that the intercept equals $H^f(T^m)$. For a concave $G^f(T)$, which is larger, $H^f(T^m)$ or $H^f(0)$? Check the answer against Table I of [F].

Answer. (a) A straight line with the slope $-H^f/k_B$ and the intercept $\ln g_X + S^f/k_B$. (b) $H^f/k_B T^m = 9.45$ and $c^{\text{eq}} = 8.7 \times 10^{-4}$. Without S^f, it would be $e^{2.4} = 11$ times smaller. (c) Concave. (d) The tangent of a concave function lies above it, so $H^f(T^m) \geq H^f(0)$. Table I gives $0.68 > 0.63$ eV (GGA) and $0.78 > 0.68$ eV (LDA), both with anharmonicity.
Related: Sections 3.3, 4.5 and 5.3 of the note. *Source:* [F], Secs. II.B.3 and II.C.1, Eq. (42), Table I.

Exercise 25: Electronic entropy and the Fermi–Dirac distribution. Electrons are fermions, so each single-particle orbital holds either 0 or 1 electron of a given spin. Non-interacting orbitals are independent.

(a) Treat one orbital of energy ϵ_i as a grand canonical system with $N_i \in \{0, 1\}$, in contact with an electron reservoir at $\mu = E_F$. What is Ξ_i? Show that the mean occupation is the Fermi–Dirac function $f_i = 1/[e^{\beta(\epsilon_i - E_F)} + 1]$.

(b) What is the Gibbs–Shannon entropy of a two-outcome distribution with the probabilities f_i and $1 - f_i$? Why does the total entropy add over orbitals? Derive Eq. (11) of [F]:

$$S^{\mathrm{el}} = -k_B \sum_i \left[(1 - f_i) \ln(1 - f_i) + f_i \ln f_i \right].$$

(c) Compare part (b) with the ideal mixing entropy of Section 7.7 of the note. What is "mixed" here?

(d) Near E_F, let the density of states be a constant D_F, counting both spins. Turn the sum into an integral with $x = \beta(\epsilon - E_F)$ and use

$$\int_{-\infty}^{\infty} dx \, [-f \ln f - (1 - f) \ln(1 - f)] = \frac{\pi^2}{3}, \qquad f = \frac{1}{e^x + 1},$$

to show that $S^{\mathrm{el}} = (\pi^2/3) k_B^2 T D_F$.

(e) The excitation energy is $\Delta E^{\mathrm{el}} = (\pi^2/6)(k_B T)^2 D_F$. Show that $\tilde{F}^{\mathrm{el}} = \Delta E^{\mathrm{el}} - T S^{\mathrm{el}} = -\frac{1}{2} T S^{\mathrm{el}}$, Eq. (13) of [F]. Why is this contribution small in a semiconductor but not in a metal?

(f) A neutral defect has one unpaired electron in a localized gap orbital that holds either spin. Which part of S^{el} survives as $T \to 0$, and where does it reappear in Exercise 23?

Answer. (a) $\Xi_i = 1 + e^{-\beta(\epsilon_i - E_F)}$ and $f_i = -\partial \ln \Xi_i / \partial(\beta \epsilon_i)$. (b) $-k_B[f \ln f + (1 - f) \ln(1 - f)]$, and independent systems have additive entropies (Section 3.2 of the note). (c) Occupied and empty orbitals are "mixed". (d) The integral is $\pi^2/3 = 3.290$. (e) $D_F \approx 0$ in the band gap of a semiconductor. (f) $k_B \ln 2$, which is the spin degeneracy $g_X = 2$ in S^{conf}.
Related: Sections 1.1, 3.2, 5 and 7.7 of the note. *Source:* [F], Sec. II.A.2, Eqs. (10)–(13).

Exercise 26: Vibrational free energy and the Bose–Einstein distribution. Near equilibrium, the energy of the ions is quadratic in the displacements $u_k = R_k - R_k^0$, Eq. (9) of [F].

(a) Newton's equations read $M_k \ddot{u}_k = -\sum_l \Phi_{kl} u_l$, with the force constants $\Phi_{kl} = \partial^2 F^{\mathrm{el}} / \partial R_k \partial R_l$. Substitute $u_k = w_k e^{-i\omega t} / \sqrt{M_k}$. Why does the dynamical matrix $D_{kl} = \Phi_{kl} / \sqrt{M_k M_l}$, Eq. (14) of [F], appear, and why are its eigenvalues ω_i^2?

(b) A quantum oscillator has the levels $(n + \frac{1}{2})\hbar\omega$, $n = 0, 1, 2, \ldots$. Sum the geometric series for Z. Show that $F = \hbar\omega/2 + k_B T \ln(1 - e^{-\beta\hbar\omega})$, one term of Eq. (17) of [F].

(c) Show that the mean energy is $(\frac{1}{2} + \bar{n})\hbar\omega$ with $\bar{n} = 1/(e^{\beta\hbar\omega} - 1)$, Eq. (16) of [F]. Read $\bar{n}$ as the mean number of phonons in the mode. Compare $\bar{n}$ with f of Exercise 25. Which sign changed, and why is the phonon chemical potential zero?

(d) Find the high-temperature limit of F. Show that the vibrational formation entropy becomes $S^f_{\rm vib} = k_B \sum_i \ln(\omega_i^{\rm bulk}/\omega_i^{\rm def})$, with equal numbers of modes on both sides.

(e) Around a vacancy, the bonds of the neighbor atoms are broken. Do their modes soften or stiffen, and what is the sign of $S^f_{\rm vib}$? Estimate $S^f_{\rm vib}$ if the 36 modes of the 12 nearest neighbors in an fcc metal soften by 5%, and compare with $S^f = 2.4\,k_B$ for Al.

Answer. (a) The substitution turns the equations into the eigenvalue problem $Dw = \omega^2 w$, Eq. (15) of [F]. D is real and symmetric, and stability makes it positive definite. (b) $Z = e^{-\beta\hbar\omega/2}/(1-e^{-\beta\hbar\omega})$. (c) -1 replaces $+1$ in the denominator, because any number of phonons can occupy a mode. Phonon numbers are not conserved. (d) $F \to k_B T \ln(\hbar\omega/k_B T)$. (e) The modes soften, and $S^f_{\rm vib} > 0$. The estimate gives $36 \ln(1/0.95) = 1.8\,k_B$.
Related: Sections 3, 3.3 and 4.5 of the note. *Source:* [F], Sec. II.A.3, Eqs. (9), (14)–(17).

Exercise 27: Formation volume and pressure. At finite T and P, the formation Gibbs energy of Eq. (23) of [F] contains the term PV^f, with the formation volume $V^f = \Omega' - \Omega$.

(a) Which derivative of $G^f(T,P)$ gives V^f? Use Section 5.3 of the note.

(b) Show that $(\partial \ln c^{\rm eq}/\partial P)_T = -V^f/k_B T$. For a vacancy in Al with $V^f = 11$ Å^3, by what factor does 1 GPa change $c^{\rm eq}$ at 900 K?

(c) A supercell of fixed volume V_0 prevents the relaxation V^f. Use $B = -V\,\partial P/\partial V$ to show that the elastic energy stored is $\Delta E \approx B(V^f)^2/(2V_0)$, Eq. (57) of [F]. Evaluate ΔE for $B = 100$ GPa, $V^f = 10$ Å^3 and $V_0 = 1000$ Å^3.

(d) How does ΔE scale with the supercell size L? Why does stability, $B > 0$, make the constant-volume energy higher than the relaxed energy?

Answer. (a) $V^f = (\partial G^f/\partial P)_T$. (b) $\Delta \ln c = -0.89$, so c drops by a factor of 2.4. (c) $\Delta E = \int_0^{V^f} B\,(V/V_0)\,dV = 31$ meV. (d) $\Delta E \propto 1/L^3$. The constrained state lies on a convex $F(V)$ away from the minimum.
Related: Sections 4.1, 4.6 and 5.3 of the note. *Source:* [F], Secs. II.B.1 and III.D, Eqs. (23), (57).

Exercise 28: Chemical potentials and the stability window. A nitrogen vacancy $V_{\rm N}$ forms in GaN, which coexists either with Ga metal or with N_2 gas.

(a) In GaN at equilibrium, why must $\mu_{\rm Ga} + \mu_{\rm N} = \mu({\rm GaN})$, Eq. (26) of [F]? Use the Euler relation $G = \sum_i \mu_i N_i$ of Section 5.5 of the note.

(b) If μ_{Ga} exceeded μ_{Ga}(Ga metal), what would form? Derive the bounds of Eqs. (27)–(30) of [F] and show that

$$\Delta_f G^0(\mathrm{GaN}) \le \Delta\mu_{\mathrm{Ga}} \le 0, \qquad \Delta\mu_{\mathrm{N}} = \Delta_f G^0(\mathrm{GaN}) - \Delta\mu_{\mathrm{Ga}},$$

in the notation of Eqs. (31)–(34).

(c) Draw $g(X)$ for the Ga–N system at fixed (T, P), with three line phases at $X = 0$ (Ga), $X = 1/2$ (GaN) and $X = 1$ (N_2), and construct the convex hull of Section 7.4 of the note. Which tangent lines touch the GaN point? Show that their intercepts at $X = 0$ and $X = 1$ sweep exactly the window of part (b).

(d) The formation of V_{N} removes one N atom, $n_{\mathrm{N}} = -1$. How much larger is $E^f(V_{\mathrm{N}})$ under N-rich than under Ga-rich conditions?

(e) For N_2 gas, the molecule has the ground energy $E(N_2)$ and the rotational partition function $z_{\mathrm{rot}} = k_B T/(\sigma B_0)$ for $k_B T \gg B_0$. Start from $\mu = k_B T \ln(P\lambda_{\mathrm{th}}^3/k_B T)$ of Exercise 13 and multiply Z_N by z_{int}^N. Derive Eq. (35) of [F]:

$$2\mu_{\mathrm{N}} = E(\mathrm{N_2}) + k_B T \ln \frac{P V_Q}{k_B T} + k_B T \ln \frac{\sigma B_0}{k_B T} + \mu_{\mathrm{vib}}, \qquad V_Q = \lambda_{\mathrm{th}}^3.$$

(f) Raising $P(N_2)$ at fixed T moves μ_{N} in which direction? How does $c(V_{\mathrm{N}})$ respond?

Answer. (b) Excess Ga would precipitate as metal, and excess N would leave as N_2. (c) The tangents through the GaN point rotate between the slopes of the Ga–GaN and GaN–N_2 segments of the hull. (d) By $|\Delta_f G^0(\mathrm{GaN})|$, about 1 eV. (e) $\mu = -k_B T \ln z_{\mathrm{int}} + k_B T \ln(P\lambda_{\mathrm{th}}^3/k_B T)$, with $z_{\mathrm{int}} = e^{-\beta E(\mathrm{N_2})} z_{\mathrm{rot}} z_{\mathrm{vib}}$. (f) μ_{N} rises by $\frac{1}{2} k_B T\, \Delta \ln P$, and $c(V_{\mathrm{N}}) \propto P^{-1/2}$.
Related: Sections 5.5, 5.7, 7.3 and 7.4 of the note. *Source:* [F], Sec. II.B.2, Eqs. (26)–(35).

Exercise 29: Charge neutrality and the Fermi level. A semiconductor has the band gap E_g, parabolic bands with the effective masses m_e and m_h, and dilute charged defects. Electrons and holes follow Exercise 25.

(a) Why must the crystal as a whole be neutral, Eq. (36) of [F]? Which unknown does this condition fix, once all $c(X^q)$ are written as functions of E_F?

(b) The conduction band has the density of states per volume

$$D_c(E) = \frac{1}{2\pi^2} \left(\frac{2m_e}{\hbar^2}\right)^{3/2} \sqrt{E - E_c},$$

with both spins included. When $E_c - E_F \gg k_B T$, which term of the Fermi–Dirac denominator can be dropped? Show that $n_e = N_c e^{-(E_c - E_F)/k_B T}$ with $N_c = 2(m_e k_B T / 2\pi\hbar^2)^{3/2}$. Compare with Eq. (38) of [F], which counts one spin.

(c) Evaluate N_c for m_e equal to the free-electron mass at 300 K.

(d) Without defects, $n_e = n_h$. Show that $E_F = E_g/2 + \frac{3}{4} k_B T \ln(m_h/m_e)$.

(e) A donor D^+ has $E^f = E_D + E_F$, and a native acceptor A^- has $E^f = E_A - E_F$, with $c \propto e^{-E^f/k_B T}$. If the defects outnumber the free carriers, where does neutrality pin E_F? Locate this point in a plot of E^f against E_F.

(f) Raising the dopant chemical potential μ_D lowers E_D by the same amount. Show that E_F then rises only by half of the increase of μ_D. Why does this effect, self-compensation, limit doping (Sec. I.A.2 of [F])?

Answer. (a) The crystal has no external source of charge, so neutrality fixes E_F. (b) $f \approx e^{-\beta(E - E_F)}$, and the integral is a Gaussian integral. (c) $N_c = 2.5 \times 10^{19}$ cm^{-3}. (d) $N_c/N_v = (m_e/m_h)^{3/2}$. (e) $c(D^+) = c(A^-)$ gives $E_F = (E_A - E_D)/2$, where the two lines cross. (f) $E_F = (E_A - E_D)/2$ rises by $\Delta\mu_D/2$. Each added donor is compensated by a new acceptor, so the carrier density barely grows. *Related:* Sections 5, 5.5 and 6.3 of the note. *Source:* [F], Secs. I.A.2 and II.B.2c, Eqs. (36)–(38).

Exercise 30: Vacancies close to the melting point. Section V.B of [F] asks whether vacancies cause the rise of the heat capacity of Al close to T^m.

(a) From Eq. (40) of [F], the vacancy contribution to the Gibbs energy per site is $\Delta G = -k_B T c^{\mathrm{eq}}(T)$. Use $H = -T^2\, \partial(G/T)/\partial T$ to show that $\Delta H = c^{\mathrm{eq}} H^f$. Why is this result expected?

(b) Show that $\Delta C_P = k_B c^{\mathrm{eq}} (H^f/k_B T)^2$.

(c) The number of vacancies in N sites is a sum of nearly independent site occupations. What is its variance? Use $\mathrm{Var}(U) = k_B T^2 C$ of Section 3.3 of the note to obtain part (b) a second way.

(d) Evaluate ΔC_P per atom for Al at T^m with the data of Exercise 24, and compare it with $3k_B$.

(e) Which other response function of Section V.B of [F] should reveal vacancies, and why does it depend on V^f?

Answer. (a) Each vacancy carries the enthalpy H^f. (c) $\mathrm{Var}(n) \approx n$, so $\mathrm{Var}(U) = (H^f)^2 n$. (d) $\Delta C_P = 0.078\, k_B$ per atom, about 2.6% of $3k_B$. (e) The thermal expansion coefficient, because each

vacancy adds V^f to the volume.
Related: Sections 3.3, 4.6 and 5.3 of the note. *Source:* [F], Secs. II.B.3 and V.B, Eq. (40).

Exercise 31: Optical levels and the configuration-coordinate diagram. In Fig. 8 of [F], the substitutional acceptor $\mathrm{N_O}$ in ZnO has the absorption energy $E_{\mathrm{abs}} = 2.4$ eV and the emission energy $E_e = 1.7$ eV. ZnO has $E_g = 3.4$ eV.

(a) Model the two states $\mathrm{N_O^-}$ and $\mathrm{N_O^0} + e^-_{\mathrm{CB}}$ by parabolas $\frac{1}{2}K(Q-Q_-)^2$ and $E_{\mathrm{ZPL}} + \frac{1}{2}K(Q-Q_0)^2$ with equal curvature K. Why do absorption and emission occur at fixed Q (Franck–Condon principle)?

(b) Show that $E_{\mathrm{abs}} = E_{\mathrm{ZPL}} + E_{\mathrm{rel}}$ and $E_e = E_{\mathrm{ZPL}} - E_{\mathrm{rel}}$, with $E_{\mathrm{rel}} = \frac{1}{2}K(Q_- - Q_0)^2$. Find E_{rel}, the Stokes shift and E_{ZPL}.

(c) The zero-phonon line equals $E_g - \varepsilon(0/-)$. Find $\varepsilon(0/-)$ and compare with 1.3 eV in [F].

(d) Why does a Fermi-level-dependent concentration use the thermodynamic level, whereas an absorption spectrum uses the optical level? Compare the time scales of electronic and ionic motion.

(e) For a typical phonon energy $\hbar\omega = 40$ meV, how many phonons does the relaxation emit? Why does this number broaden the emission band?

Answer. (b) $E_{\mathrm{rel}} = 0.35$ eV, the Stokes shift is 0.7 eV, and $E_{\mathrm{ZPL}} = 2.05$ eV. (c) $\varepsilon(0/-) = 1.35$ eV. (d) Electronic transitions take about 10^{-15} s, ionic relaxation about 10^{-13} s. (e) About $E_{\mathrm{rel}}/\hbar\omega \approx 9$ phonons.
Related: Sections 3 and 6 of the note. *Source:* [F], Sec. II.E, Fig. 8, Eq. (43).

Exercise 32: Diffusion by vacancy hopping. Section I.A.3 of [F] states that impurities usually diffuse with the help of vacancies or interstitials. This exercise builds the diffusion coefficient from the canonical ensemble.

(a) An atom makes uncorrelated jumps of length a in random directions, Γ jumps per unit time. Why do the cross terms $\langle \vec{a}_j \cdot \vec{a}_k \rangle$, $j \neq k$, vanish? Show that $\langle r^2 \rangle = \Gamma a^2 t$, and define D by $\langle r^2 \rangle = 6Dt$.

(b) Check that $c(\vec{r}, t) \propto t^{-3/2} e^{-r^2/4Dt}$ solves $\partial c/\partial t = D\nabla^2 c$ and has $\langle r^2 \rangle = 6Dt$.

(c) In one dimension, an atom sits in a well of curvature K and mass m, so that the attempt frequency is $\nu = \sqrt{K/m}/2\pi$, and a barrier of height E_m separates the well from the next one. In canonical equilibrium, what is the probability density of finding the atom at the barrier top, normalized over the well? What is the mean forward velocity $\langle v\,\theta(v)\rangle$ from the Maxwell distribution? Multiply the two to show that the jump rate is $\Gamma_0 = \nu e^{-E_m/k_BT}$, the result of transition-state theory.

(d) An atom in a crystal jumps only into a neighboring vacancy. With z neighbors and the vacancy concentration c_v^{eq} of Exercise 23, show that the self-diffusion coefficient is

$$D = \frac{f_c\, z\, a^2 \nu}{6} \exp\!\left(\frac{S^f}{k_B}\right) \exp\!\left(-\frac{H^f + E_m}{k_B T}\right),$$

where the correlation factor f_c (0.78 for fcc) corrects for jumps straight back into the same vacancy.

(e) Which energy does an Arrhenius plot of D measure? For Al, with $H^f = 0.67$ eV and $E_m = 0.61$ eV, compare with the measured activation energy of about 1.3 eV.

(f) Section I.A.3 of [F] says that interstitials move more readily than vacancies but are less abundant. Which sum decides the mechanism that dominates diffusion?

(g) In a non-uniform crystal, the flux is $J = -(Dc/k_BT)\,\partial\mu/\partial x$. For a dilute species with $\mu = k_BT \ln c + \text{const}$, recover Fick's law. Why does this result agree with the statement of Section 7.11 of the note that no flux flows through a two-phase zone?

Answer. (a) Independent jumps have zero mean. (c) The density at the barrier top, normalized over the well, is $\sqrt{K/2\pi k_BT}\, e^{-\beta E_m}$, and $\langle v\,\theta(v)\rangle = \sqrt{k_BT/2\pi m}$. (e) $H^f + E_m = 1.28$ eV. (f) $H^f + E_m$ for each mechanism. (g) $J = -D\,\partial c/\partial x$. Inside a two-phase zone μ is uniform, so $J = 0$. *Related:* Sections 3, 7.4 and 7.11 of the note. *Source:* [F], Secs. I.A.3 and II.B.3, Eq. (41).

Exercise 33: Charged defects in a periodic supercell. A supercell calculation repeats one charged defect q on a cubic lattice of period L in a medium of dielectric constant ε.

(a) Why does the electrostatic energy of a periodic array of net charges diverge? Why does a uniform compensating background remove the divergence? What happens to the $\vec{G} = 0$ Fourier component of the potential?

(b) On dimensional grounds, how must the spurious interaction scale with q, ε and L? For a simple cubic array of point charges in a background, the Madelung energy is $E_M = \alpha_M q^2/(2\varepsilon L)$ with $\alpha_M = 2.8373$. Evaluate E_M for $q = 1$, $\varepsilon = 10$ and $L = 12$ Å.

(c) The computational cost grows as the cube of the number of atoms. By what factor does the cost grow if the error is halved? This is the estimate of Sec. I.C.2 of [F].

(d) The energy behaves as $E(L) = E_\infty + a_1/L + a_3/L^3$, with an L^{-3} term from the quadrupole moment. How many supercell sizes determine E_∞? How many if a_1 is taken from the Madelung energy?

(e) Why is the Fermi level of Exercise 21 undefined in a charged supercell until the average electrostatic potential is aligned with that of the bulk (Fig. 12 of [F])?

Answer. (a) The Coulomb sum over the net charges grows with the number of cells. The background sets the average charge, the $\vec{G} = 0$ term, to zero. (b) $E_M \propto q^2/(\varepsilon L)$, and $E_M = 0.17$ eV, larger than many target accuracies. (c) Halving the error doubles L, so the number of atoms grows 8 times and the cost $8^3 = 512$ times. (d) Three sizes, or two with a_1 known. (e) E^f contains qE_F, and E_F is measured from the VBM, whose absolute energy depends on the reference of the potential.
Related: Section 5.2 of the note. *Source:* [F], Secs. I.C.2 and III.C.